Supply Chain & Delivery Performance

Year: All | Market: All | Shipping Mode: All

Total Sales
$36.78M

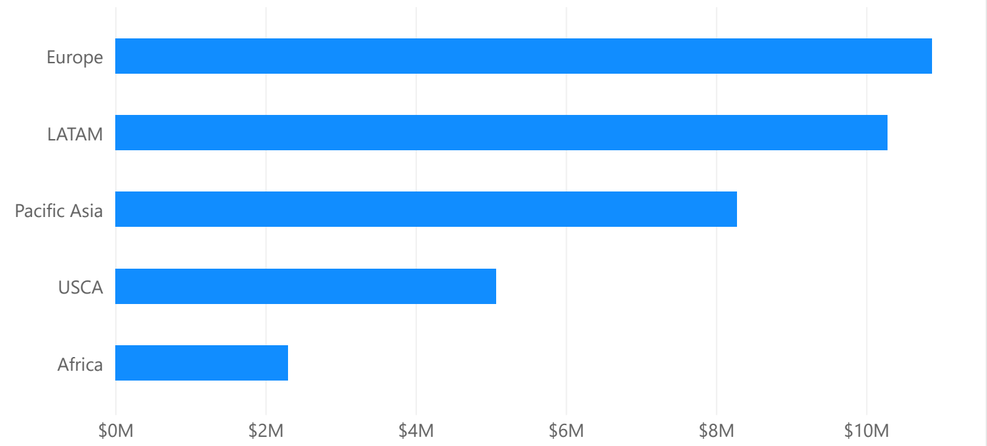
Total Profit
$3.97M

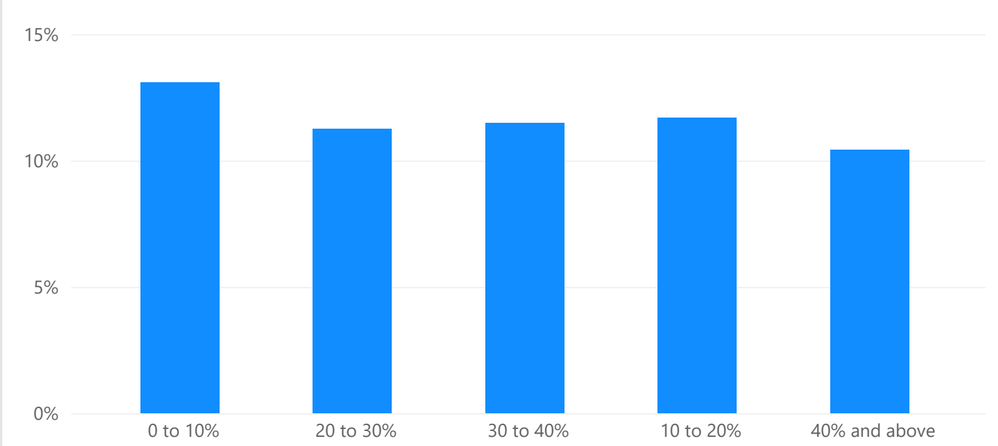
Profit Margin %
10.78%

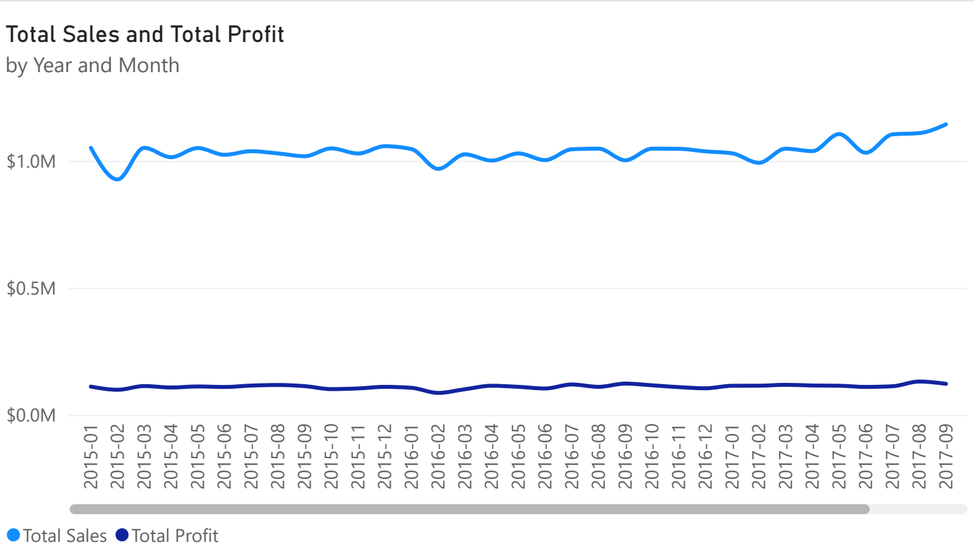
Total Orders
66K

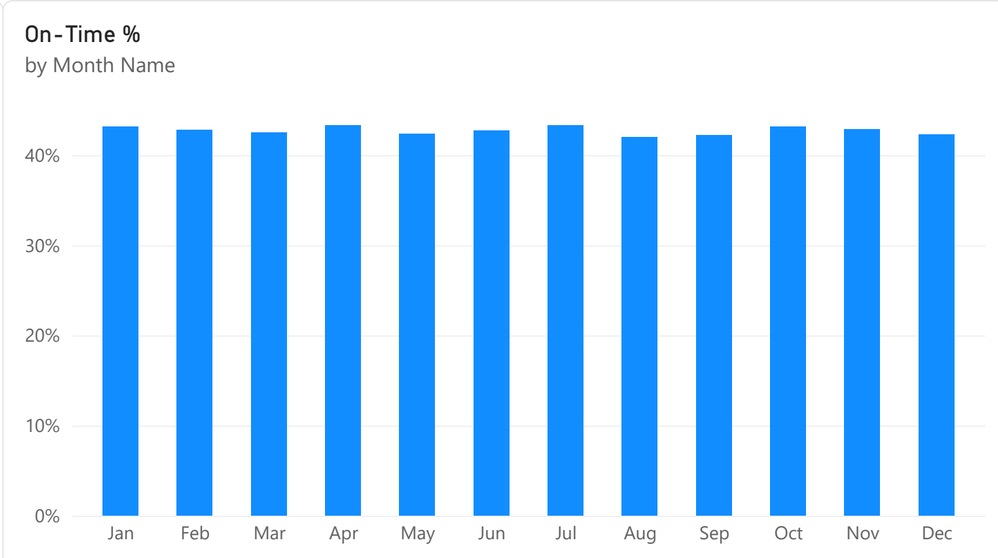
On-Time %
42.71%

Loss as % of Total Profit
0.04%





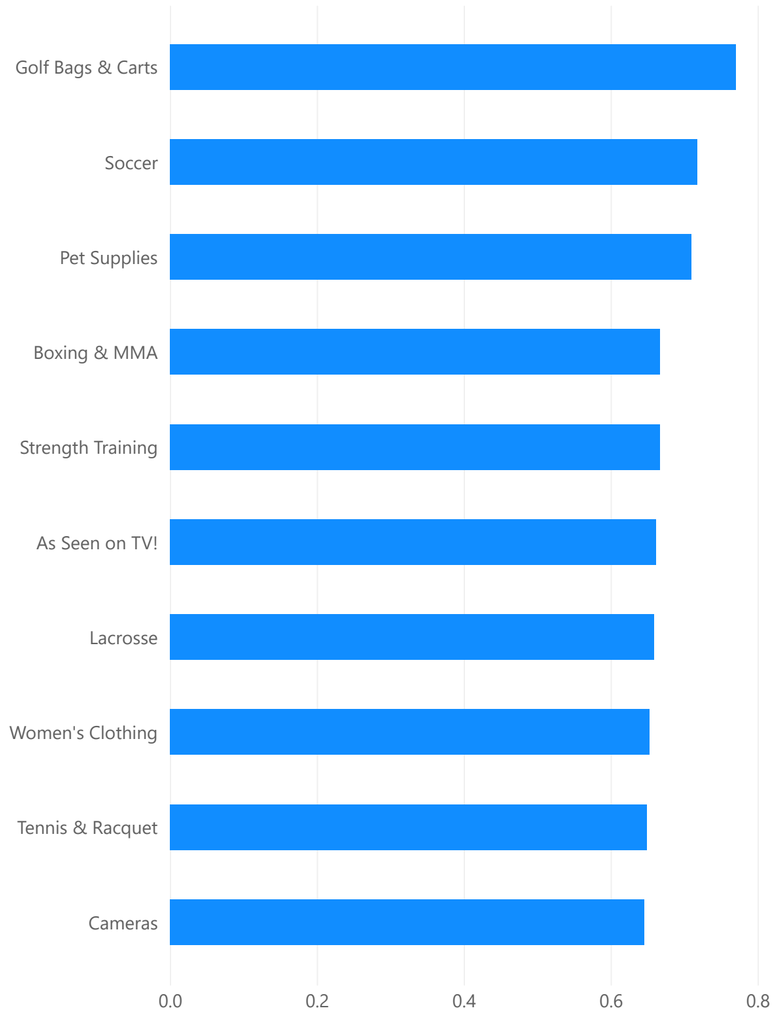




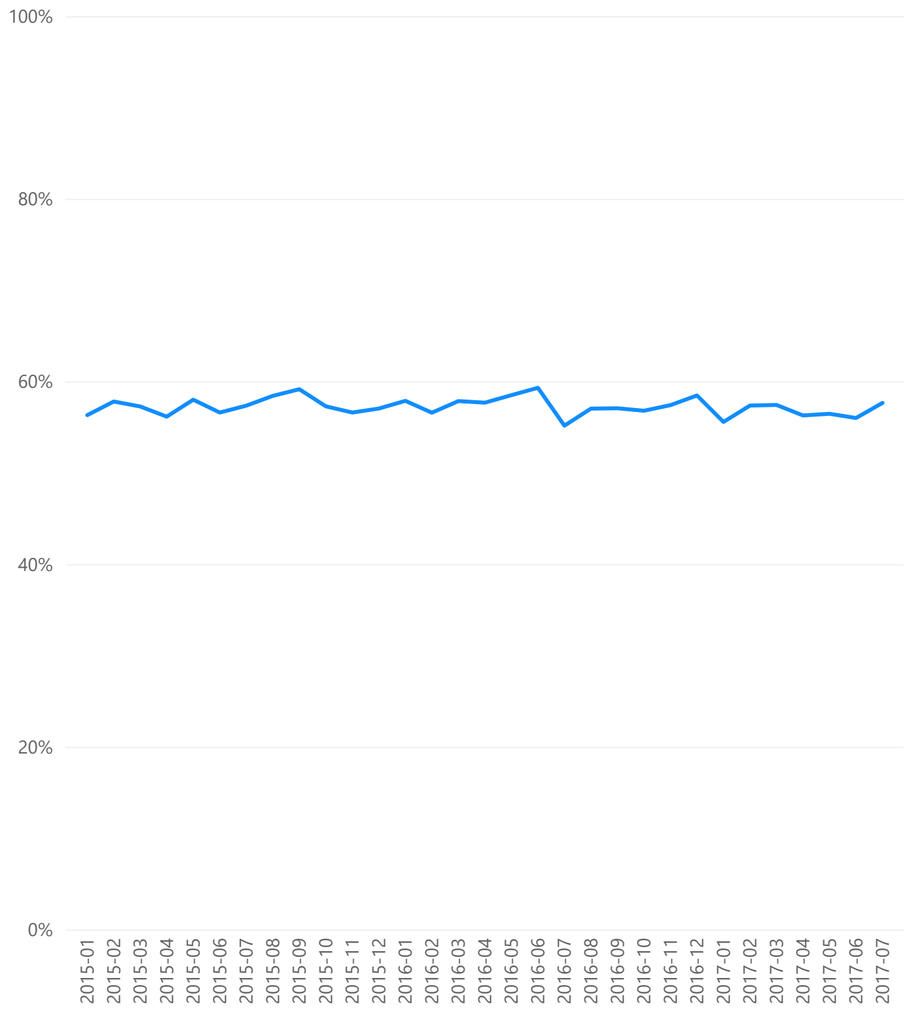
Delivery Performance

Year: All | Market: All | Shipping Mode: All

Avg Shipping Delay
by Category



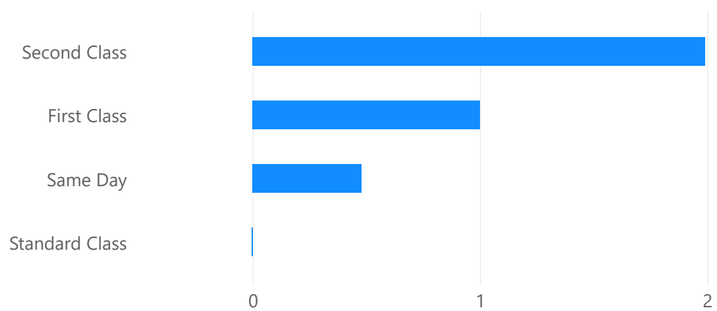
Late Delivery %
by Year Month



On-Time %
by Region & Shipping Mode

Order Region	First Class	Same Day	Second Class	Standard Class
Canada	0.00%	75.86%	18.83%	70.04%
West Africa	0.00%	62.74%	20.61%	61.11%
Caribbean	0.00%	47.66%	17.51%	63.20%
Southern Africa	0.00%	33.33%	26.62%	60.46%
Northern Europe	0.00%	58.96%	20.76%	60.54%
Oceania	0.00%	56.96%	20.51%	59.45%
Central Asia	0.00%	79.69%	9.82%	57.43%
Eastern Asia	0.00%	50.64%	16.78%	61.03%
West of USA	0.00%	50.12%	19.82%	60.82%
Southern Europe	0.00%	43.36%	20.67%	61.67%
North Africa	0.00%	41.62%	18.14%	61.99%
Total	**0.00%**	**52.07%**	**20.17%**	**60.23%**

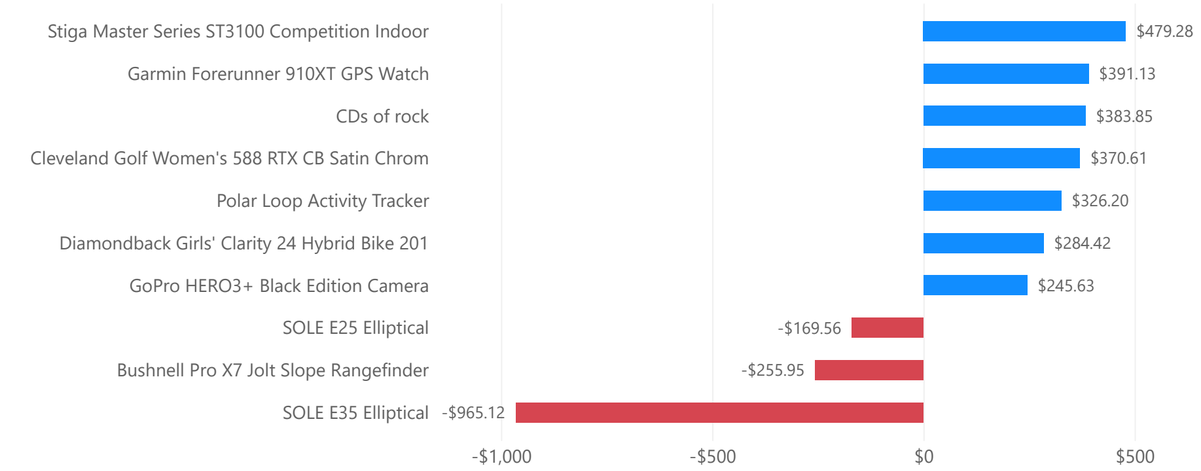
Avg Shipping Delay (Days)
by Shipping Mode



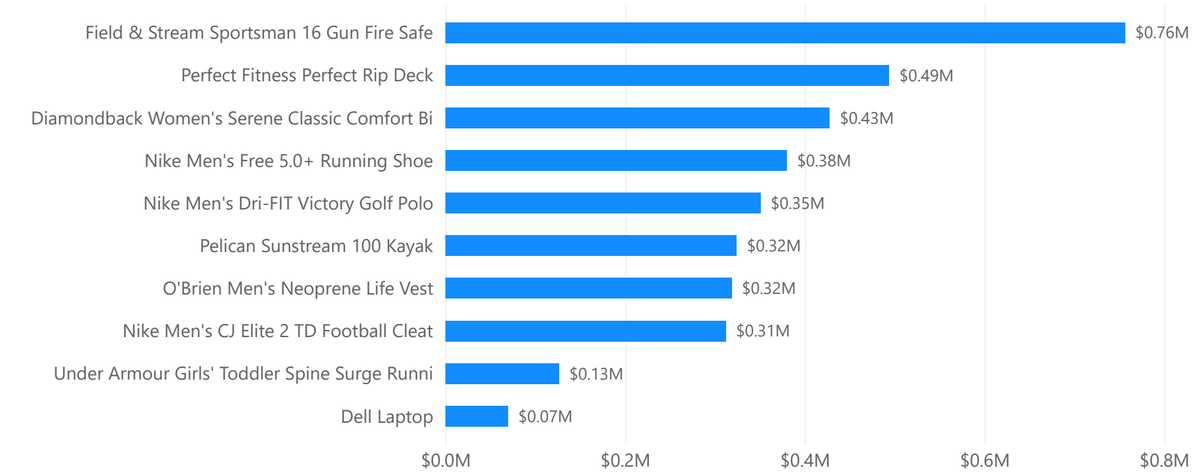
Profitability & Discount Impact Analysis

Year: All | Market: All | Shipping Mode: All

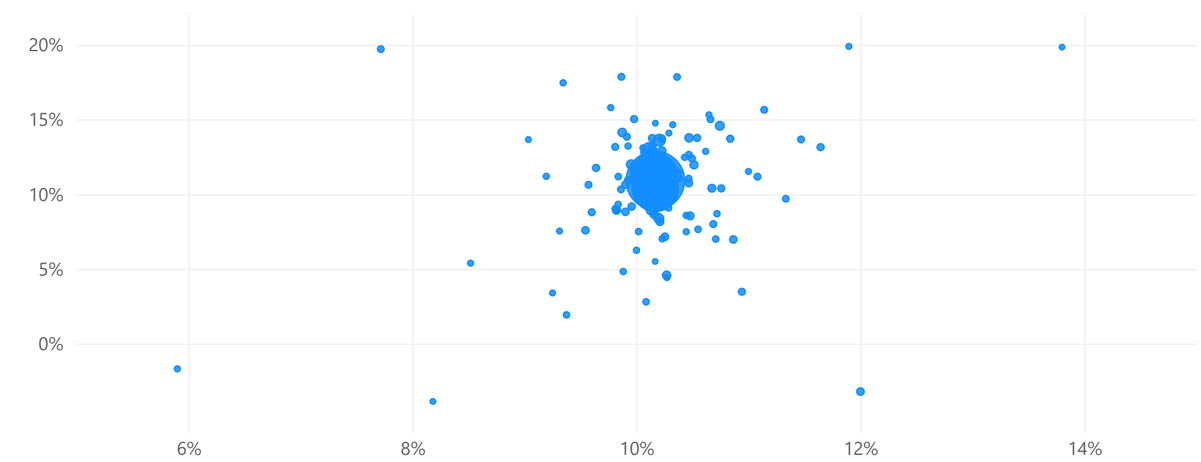
Least Profitable Products



Most Profitable Products

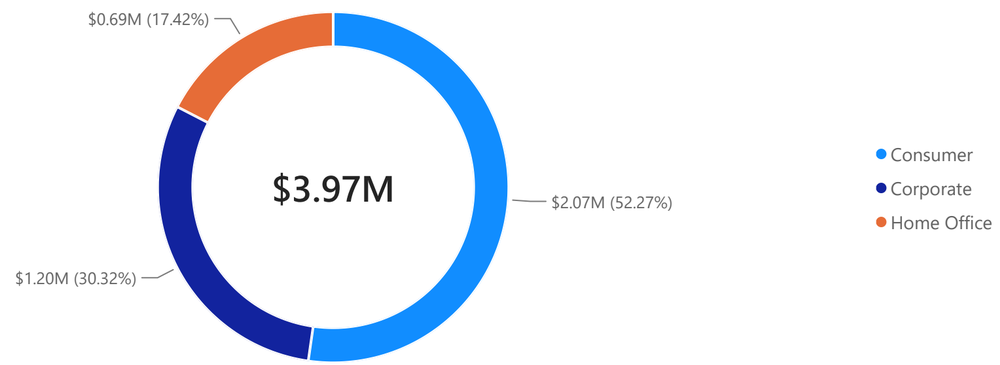


Discount Rate vs. Profit Margin



Profit Share

by Customer Segment



Executive Insights & Action Plan

Key Operational Findings

1. Shipping SLA Bottleneck
   - *Finding: First Class shipping represents a severe fulfillment failure, achieving an on-time delivery rate of only 0.0% and averaging ~1 day of delay across all product categories.*
   - *Visual Reference: Delivery Performance -> On-Time % by Region & Shipping Mode Matrix.*

2. Catalog Loss Concentration (Profitability Analysis Page)
   - *Finding Only 3 products loose money, totaling about $1.4K, under 0.04% of total profit.*
   - *Visual Reference:* Profitability Analysis -> *Bottom 10 Products by Profit Bar Chart.*

3. Discount Rate Margin Erosion
   - *Finding: Higher discount levels are associated with lower profit margins. While discount bands below 40% maintain margins around 11% to 13%, orders discounted by 40% or more deliver the weakest profitability, highlighting the need for tighter controls on deep discounting strategies.*
   - *Visual Reference:* Profitability Analysis -> *Discount Rate vs. Profit Margin Scatter Plot.*

4. Core Revenue & Profit Driver (Executive Overview Page)
   - *Finding: The Consumer customer segment is the primary financial pillar, contributing 52.27% ($2.07M) of total net profit across 66K completed orders.*
   - *Visual Reference: Executive Overview -> Top KPI Cards & Profitability Analysis -> Profit Share Donut Chart.*

5. Regional SLA (Delivery Performance Page)
   - *Finding: Regional fulfillment exhibits low variance—Western Europe (41.47% on-time) and Central Africa (39.85% on-time) have the least on-time %.*
   - *Visual Reference: Delivery Performance -> On-Time % Matrix.*

Strategic Recommendations

- **Renegotiate First Class Carrier SLAs:**
  Transition high-priority orders away from First Class shipping or renegotiate carrier contracts in low-performing regions (*Western Europe*, *Central Africa*).

- **Delist / Reprice Loss-Making Products:**
  Halt automated discounting or renegotiate supplier procurement costs for the 3 negative-margin fitness items (*SOLE E35*, *SOLE E25*, *Bushnell Pro*) to eliminate bottom-line losses.

- **Enforce Automated Discount Guardrails:**
  Implement mandatory managerial approval in order processing for any discount rate exceeding 40**%** to safeguard overall margin percentages.

Project & Dataset Overview

Project & Dataset Overview:

- **Data Source:** Kaggle DataCo Supply Chain Dataset (**180,000+ order lines**, **$36.78M total sales**, **53K orders**).
- **Environment:** Power BI Desktop (Windows), Power Query (M Language), DAX.
- **Architecture:** 1 Fact table (Fact_Orders) connected to 5 Dimension tables (Dim_Customers, Dim_Products, Dim_Geography, Dim_ShippingMode, Dim_Date) in a Star Schema model.

Business Rules & Key Metric Dictionary

- **Total Sales**
  - **Logic:** SUM(Fact_Orders[Sales])
  - **Definition:** Gross revenue generated across all valid (non-canceled) order lines.
- **Total Profit**
  - **Logic:** SUM(Fact_Orders[Order Profit Per Order])
  - **Definition:** Net profit accumulated after product manufacturing costs and discounts.
- **Profit Margin %**
  - **Logic:** [Total Profit] / [Total Sales]
  - **Definition:** Percentage of gross sales revenue retained as net profit.
- **On-Time %**
  - **Logic:** On-Time Orders / Total Valid Deliveries
  - **Definition:** Share of orders delivered where Actual Shipping Days <= Scheduled Days (excludes canceled shipments).
- **Late Delivery %**
  - **Logic:** Late Orders / Total Valid Deliveries
  - **Definition:** Share of completed orders that exceeded their scheduled shipment window.
- **Avg Shipping Delay (Days)**
  - **Logic:** AVERAGE(Fact_Orders[Days for shipping] - Fact_Orders[Scheduled days])
  - **Definition:** Average number of days orders were delayed past promised carrier SLAs.